

GPUs and Cooling Systems Assignment

1. Identify the GPU Installed

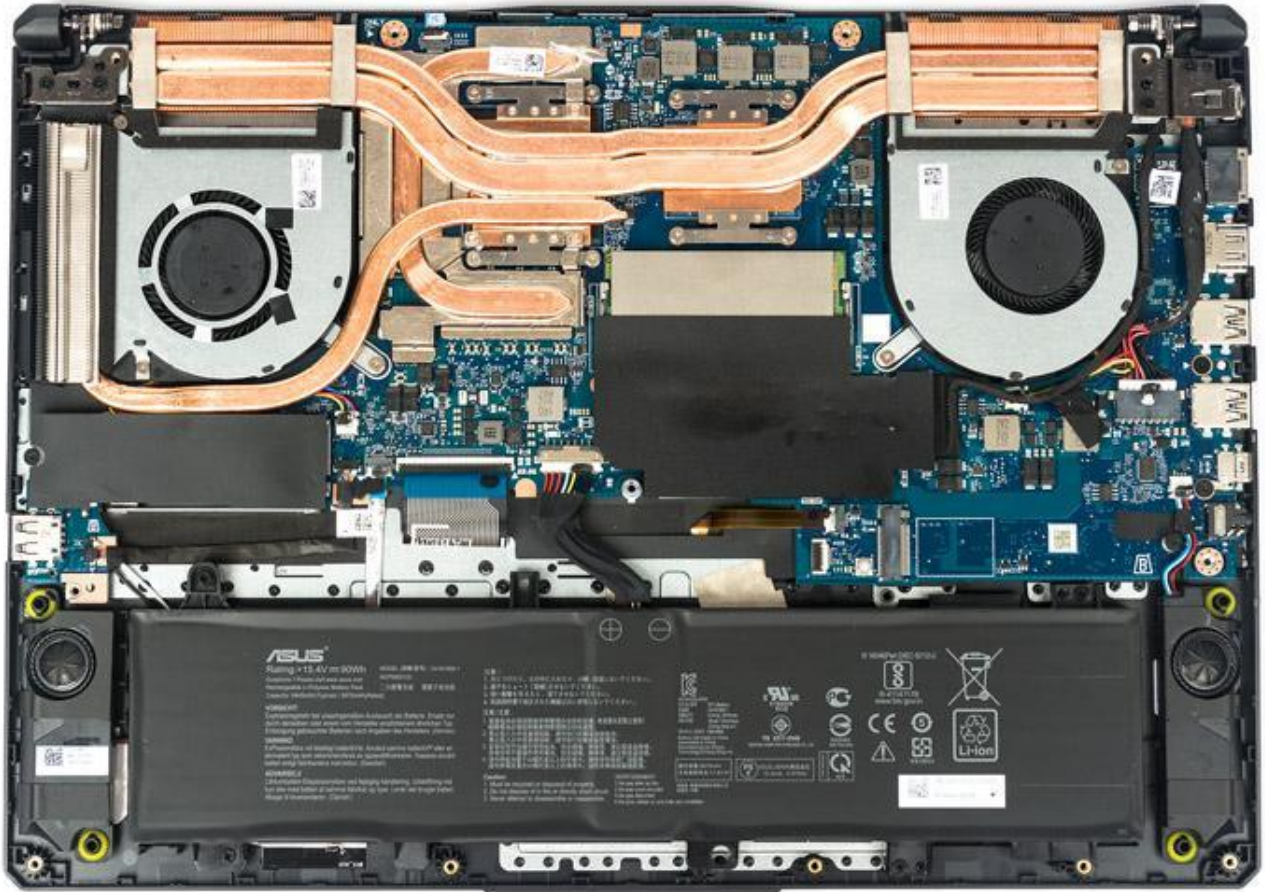
Since you mentioned you have an **ASUS TUF A15**, many models come with dedicated NVIDIA graphics. You should verify your exact GPU by pressing **Windows + X → Device Manager → Display Adapters**.

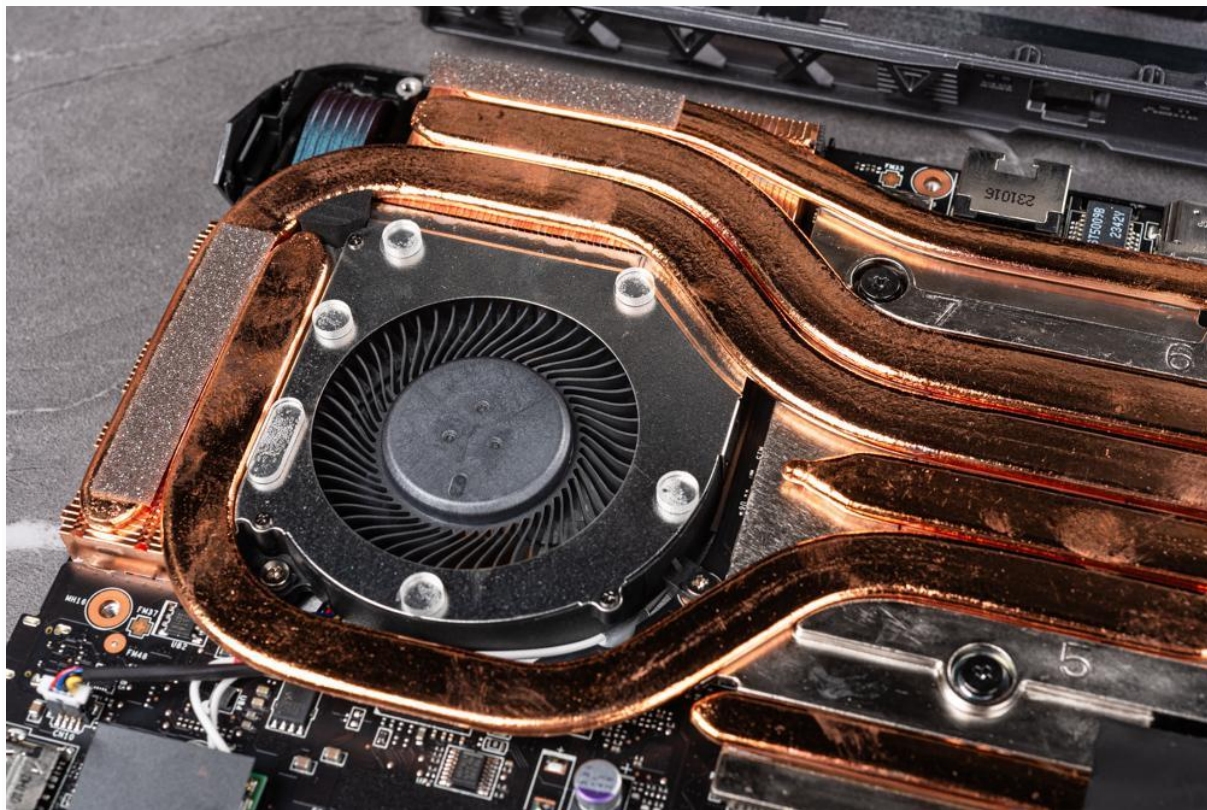
Example (common ASUS TUF A15 configuration)

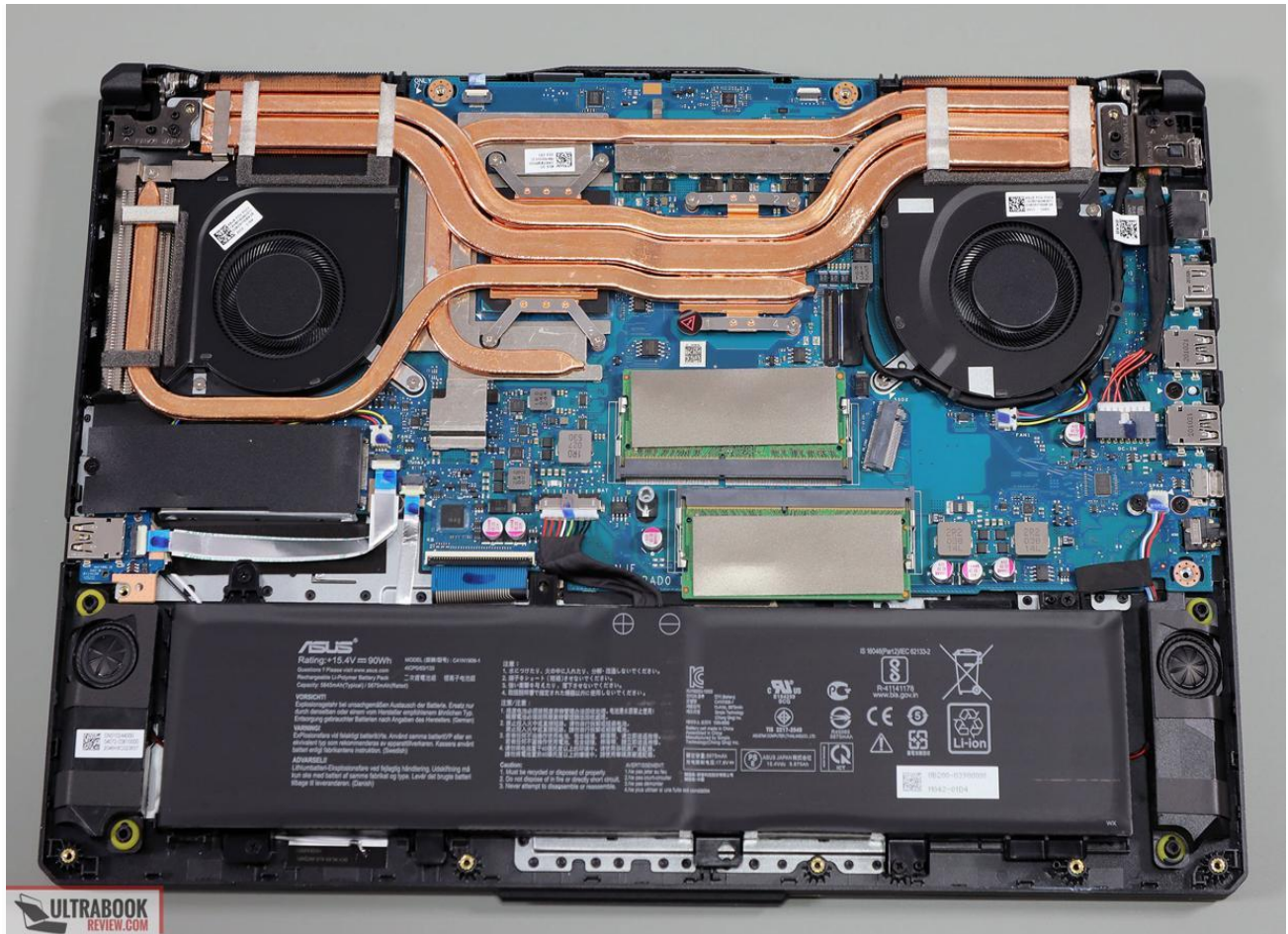
Component	Details
GPU Brand	NVIDIA
GPU Model	GeForce RTX 3050 Laptop GPU
Type	Dedicated GPU
Integrated GPU AMD Radeon Graphics (built into Ryzen processor)	

2. Cooling System Identification

Reference Cooling System Image







Label	Component	Function
1	Cooling Fan (Left)	Pulls hot air away from internal components.
2	Cooling Fan (Right)	Improves airflow and heat removal.
3	Heat Pipes	Transfer heat from CPU and GPU to heatsinks.
4	Heatsink Fins	Dissipate heat into surrounding air.
5	Air Vents	Allow hot air to exit the laptop.

Description

The cooling system uses heat pipes to move heat away from the CPU and GPU. Cooling fans blow air through the heatsink fins, allowing heat to escape through the vents. This helps maintain safe operating temperatures during gaming and streaming.

3. Air Cooling vs Liquid Cooling Comparison

Feature	Air Cooling	Liquid Cooling
Cost	Lower	Higher
Cooling Performance	Good	Excellent
Noise	Can be louder under load	Usually quieter
Installation	Easier	More complex
Maintenance	Minimal	Requires periodic maintenance
Reliability	Very reliable	More parts that can fail
Best Use	Most gaming PCs	High-end gaming/workstations

Key Differences

1. Liquid cooling generally provides better cooling performance.
2. Air cooling is less expensive and easier to install.
3. Liquid cooling requires more maintenance and has more components.

4. Cooling Choice for a Gaming & Streaming PC

Recommended Choice: Air Cooling

For a gaming PC used to stream IPL matches and play graphics-heavy games, I would choose air cooling. Modern air coolers provide excellent cooling performance while being more affordable and easier to maintain than liquid cooling systems. Air cooling is highly reliable because it has fewer components that can fail. For most gamers and streamers, a quality air cooler offers a good balance of performance, cost, and reliability.

Alternative View

If the system uses a very high-end GPU such as an RTX 4080 or RTX 4090 and runs for long gaming sessions, liquid cooling may be preferred because it can keep temperatures lower and reduce noise levels.